

SenNet LW30 / MB30 Meter

Communications via LoRaWAN or Modbus RTU, depending on the project.

The SenNet LW30/MB30 Meter is a DIN-rail three-phase power analyzer designed for integration into modern energy monitoring architectures. Available in two communication versions depending on project requirements: LoRaWAN (EU868 band) for installations without data cabling, or Modbus RTU over RS-485 for direct integration into existing BMS and SCADA platforms.

Class 0.5S Accuracy

Active energy measurement according to IEC 62053-21/22.

Two models with different communications

LoRaWAN EU868 or Modbus RTU RS-485, depending on the model.

DIN-rail mounting

Compact 72 mm format, installation in an electrical panel.



Two models under order

SenNet LW30 Meter

LORAWAN

Long-range, low-power wireless connectivity. Ideal when no data cabling is available.

SenNet MB30 Meter

MODBUS RTU · RS-485

Direct wired integration with existing BMS, SCADA or SenNet controllers in the installation.

The SenNet LW30/MB30 Meter current analyzer measures from 100 A to 5000 A using open-core current transformers, both rigid and flexible.

Part of the SenNet family

Compatible with all datalogger and controller models.

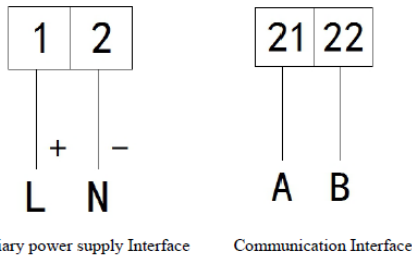
www.satel-iberia.com

Technical specifications

PHYSICAL CHARACTERISTICS		
Dimensions	720x926x715 mm width x height x depth	
Display	LCD screen with 4 multifunction buttons	
CURRENT INPUT		
Input current	External open-core 333 mV CTs or Rogowski loop-type probe	
Wiring	Summing of CTs, in 100 A steps (only for 333 mV probes)	
Consumption	< 1 VA (per phase)	
FUNCTIONAL CHARACTERISTICS		
Rated voltage	3×57.7/100 V · 3×220/380 V · 3×380/660 V · 3×100 V · 3×380 V · 3×660 V	
Consumption	< 0.5 VA (per phase)	
Reference frequency	45-65 Hz	
Available measurement	V, I, P, Q, S, PF, F, THDv, THDi, harmonics from 2nd to 31st	
LoRaWAN module	L-LRNWB25-84DN4	
AUXILIARY POWER SUPPLY		
Supply voltage	AC 85 ~ 465 V	
Consumption	< 2 W	
MEASUREMENT PERFORMANCE		
Standards	IEC 62053-22:2003 · IEC 62053-21:2003	
Active energy accuracy	Class 0.5S	
PULSE OUTPUT		
Pulse width	80 ± 20 ms	
Pulse constant	6400 imp/kWh · 400 imp/kWh	
COMMUNICATION		
SenNet LW30 Meter	LoRaWAN EU868 MHz (EU863 – 870) LoRaWAN RU868 MHz LoRaWAN AS923-1/2/3/4 LoRaWAN US915 LoRaWAN AU915	Transmit power: 16 dBm.
SenNet MB30 Meter	Modbus RTU over RS-485 (shielded twisted pair)	
ENVIRONMENTAL CONDITIONS		
Operating temperature	-20 °C ~ 60 °C	
Storage temperature	-40 °C ~ 70 °C	
Relative humidity	≤ 95 % (non-condensing)	

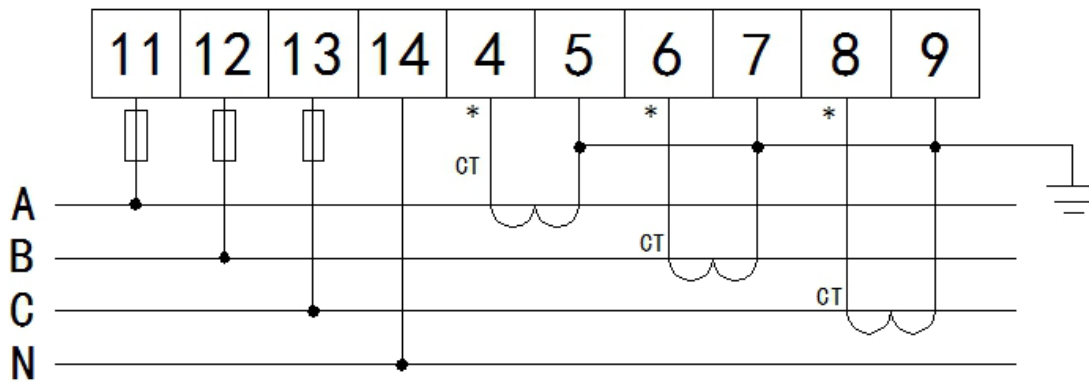
Connection diagrams

To connect the device, first connect the power supply and data lines in the case of the MB30:

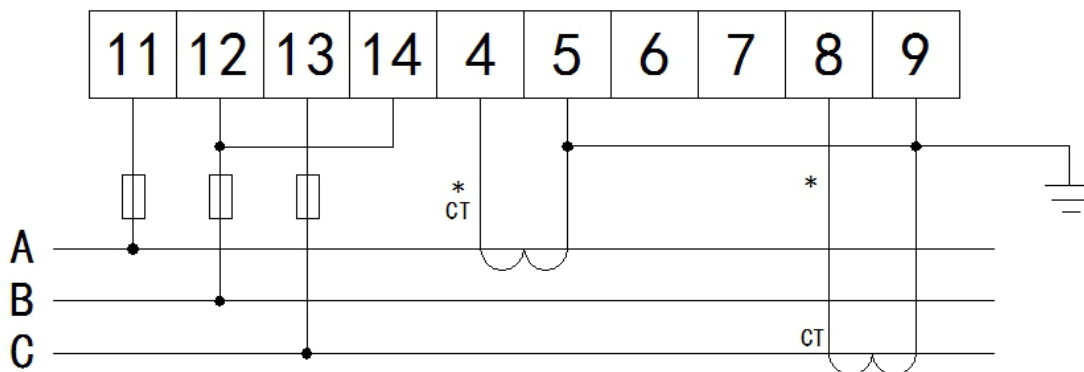


Then connect the voltage reference and the current transformers. This connection can be made with or without neutral wire, as follows:

- Voltage reference and current transformer wiring for a 4-wire three-phase line:



- Voltage reference and current transformer wiring for a 3-wire three-phase line:



Navegación mediante pantalla

The device consists of a screen and keyboard for quick and easy setup. The keys and their functions are detailed below:

TECLA	FUNCIÓN
SET	Enter settings / Return to previous menu
←	Move the position or select the digit to modify
↑	Change the value or navigate through the menu options
↵	Enter a menu, edit a parameter, or confirm a value

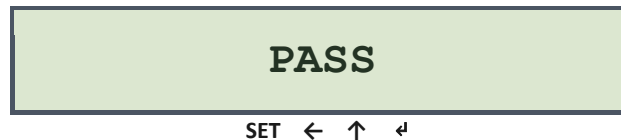
Using the keys, we will access the settings menu which allows us the following options:

PASSWORD 0000

BUS	
ADDR	Communication address
BAUD	Communication baud rate
PAR	Parity
STOP	Stop bits
SYS	
PL	Wiring system: 3P4L / 3P3L
In	
PT	Voltage transformation ratio
CT	Current transformation ratio
SAVE	Save changes: YES / NO

1. Accessing the programming mode

From the normal measurement screen, press SET:



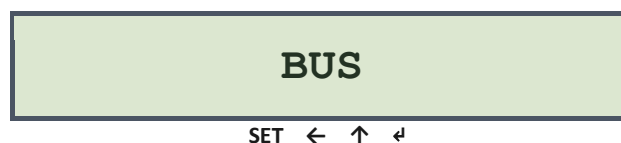
2. Enter password

Press ↵. Use ↑ to change the value and ← to move between digits. The default password is 0000.



3. Confirm password

Press ↵ to confirm. The device enters the first configuration section.



4. Communication settings - BUS

This section allows you to configure the device serial communication parameters.

Enter BUS

With BUS displayed, press ↵.

ADDR

SET ← ↑ ↵

Modbus address

Press ↵ to edit. Use ↑ y ← to set the address, then press ↵ to confirm.

ADDR 001

SET ← ↑ ↵

Note: Address range: 1 a 247.

Communication baud rate

Press ↑ to move to BAUD. Press ↵, select the baud rate with ↑ and confirm with ↵.

BAUD 9600

SET ← ↑ ↵

Parity

Press ↑ to move to PAR. Press ↵, select the parity and confirm.

PAR NONE

SET ← ↑ ↵

Note: Options: NONE, EVEN u ODD.

Stop bits

Press ↑ to move to STOP. Press ↵, select the value and confirm.

STOP 1

SET ← ↑ ↵

Nota: Options: 1 o 2.

5. Wiring System configuration - SYS

From BUS, use ↑ until SYS is displayed, then press ↵.

Select SYS

Navigate to SYS and press ↵.

SYS

SET ← ↑ ↵

Enter PL

The PL parameter defines the electrical wiring System. Press ↵.

PL

SET ← ↑ ↵

Select wiring system

Press ↵ to edit and use ↑ to select the wiring system. Confirm with ↵.

3P4L

SET ← ↑ ↵

Note: 3P4L = three-phase, four-wire. 3P3L = three-phase, three-wire.

3P3L

SET ← ↑ ↵

6. Transformation ratios – In

Use ↑ until In is displayed. This section contains the PT and CT ratios used to scale the measurements.

Enter In

With In displayed, press ↵.

In

SET ← ↑ ↵

PT Ratio

With PT selected, press ↵. Change the value using ↑ and ←, then confirm with ↵.

PT

1

SET ← ↑ ↵

Note: For direct connection, example uses PT = 1. Para un PT 10 kV/100 V, PT = 100.

CT Ratio

Press ↑ to display CT. Press ↵, change the value, and confirm.

CT

30

SET ← ↑ ↵

Note: Use 100 A, so CT = 100 = 1.

7. Saving the configuration

Changes must be saved before exiting the configuration menu.

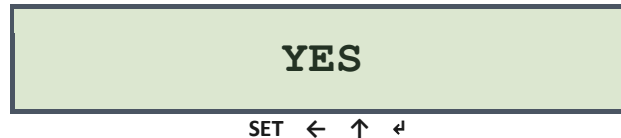
Go to SAVE

From a configuration parameter, press SET twice until the save screen is displayed.



Select YES

Press ↵ and use ↑ to select YES.



Confirm

Press ↵ to save the changes and finish.



8. Quick sequences

Change Modbus ID: SET → PASS → 0000 → ENTER → BUS → ENTER → ADDR → ENTER → value → ENTER → SET ×2 → SAVE → YES → ENTER

Change baudrate: SET → PASS → 0000 → ENTER → BUS → ENTER → ↑ BAUD → ENTER → value → ENTER → SET ×2 → SAVE → YES → ENTER

Change parity: SET → PASS → 0000 → ENTER → BUS → ENTER → ↑ hasta PAR → ENTER → value → ENTER → SET ×2 → SAVE → YES → ENTER

Change stop bits: SET → PASS → 0000 → ENTER → BUS → ENTER → ↑ hasta STOP → ENTER → value → ENTER → SET ×2 → SAVE → YES → ENTER

Change PT: SET → PASS → 0000 → ENTER → ↑ hasta In → ENTER → PT → ENTER → value → ENTER → SET ×2 → SAVE → YES → ENTER

Change CT: SET → PASS → 0000 → ENTER → ↑ hasta In → ENTER → ↑ CT → ENTER → value → ENTER → SET ×2 → SAVE → YES → ENTER

Change 3P4L / 3P3L: SET → PASS → 0000 → ENTER → ↑ SYS → ENTER → PL → ENTER → 3P4L/3P3L → ENTER → SET ×2 → SAVE → YES → ENTER

Modbus RTU communications

The following table shows the Modbus addressing of the MB30 analyzer variables for device configuration; these parameters can also be changed using the analyzer display:

ADDRESS	VARIABLE	LENGTH	FORMAT	R/W	NOTE
0	Modbus address	1	Signed	R/W	1-247
1	Speed	1	Signed	R/W	1: 1200 bps 2: 2400 bps 3: 4800 bps 4: 9600 bps
4	Parity	1	Signed	R/W	High byte: 0: none 1: even 2: odd Low byte: 0: 1 stop bit 1: 2 stop bit
7	Display sleep mode	1	Signed	R/W	Time in seconds
8	Password	1	Signed	R/W	
15	Current probe.	1	Signed	R/W	Multiplier of 100. Example: 1 = 100A 2 = 200A ... 8 = 800A
17-19	Time	1	Signed	R/W	Date, time (second, minute, hour, day, month, year)

The following table shows the Modbus addressing of the MB30 analyzer variables for reading parameters:

ADDRESS	VARIABLE	LENGTH	FORMAT	R/W	UNIT
32768	Phase 1 voltage	2	Float	R	V
32770	Phase 2 voltage	2	Float	R	
32772	Phase 3 voltage	2	Float	R	
32774	Phase 1-2 voltage	2	Float	R	
32776	Phase 2-3 voltage	2	Float	R	
32778	Phase 3-1 voltage	2	Float	R	A
32780	Phase 1 current	2	Float	R	
32782	Phase 2 current	2	Float	R	
32784	Phase 3 current	2	Float	R	kW
32788	Phase 1 active power	2	Float	R	
32790	Phase 2 active power	2	Float	R	
32792	Phase 3 active power	2	Float	R	
32794	Total active power	2	Float	R	kVar
32796	Phase 1 reactive power	2	Float	R	
32798	Phase 2 reactive power	2	Float	R	
32800	Phase 3 reactive power	2	Float	R	
32802	Total reactive power	2	Float	R	kVA
32804	Phase 1 apparent power	2	Float	R	
32806	Phase 2 apparent power	2	Float	R	
32808	Phase 3 apparent power	2	Float	R	
32810	Total apparent power	2	Float	R	Hz
32812	Phase 1 power factor	2	Float	R	
32814	Phase 2 power factor	2	Float	R	
32816	Phase 3 power factor	2	Float	R	
32818	Total power factor	2	Float	R	kWh
32820	Frequency	2	Float	R	
34942	Total active energy	2	Float	R	
34944	Active energy consumed	2	Float	R	
34946	Active energy generated	2	Float	R	kVarh
34948	Total reactive energy	2	Float	R	
34950	Reactive energy consumed	2	Float	R	
34952	Reactive energy generated	2	Float	R	

The following table shows the Modbus addressing of the MB30 analyzer variables for reading total and individual harmonics from 2nd to 31st:

ADDRESS	VARIABLE	LENGTH	FORMAT	R/W	FACTOR
116	THDV phase 1	1	Unsigned	R	0.01
117	THDV phase 2	1	Unsigned	R	
118	THDV phase 3	1	Unsigned	R	
119	THDI phase 1	1	Unsigned	R	
120	THDI phase 2	1	Unsigned	R	
121	THDI phase 3	1	Unsigned	R	
122-151	Phase 1 voltage harmonic, 2nd to 31st	1	Unsigned	R	
152-181	Phase 2 voltage harmonic, 2nd to 31st	1	Unsigned	R	
182-211	Phase 3 voltage harmonic, 2nd to 31st	1	Unsigned	R	
212-241	Phase 1 current harmonic, 2nd to 31st	1	Unsigned	R	
242-271	Phase 2 current harmonic, 2nd to 31st	1	Unsigned	R	
272-301	Phase 3 current harmonic, 2nd to 31st	1	Unsigned	R	

LoRaWAN communications

The following table shows the LoRaWAN addressing of the LW30 analyzer variables for device configuration; these parameters can also be changed using the analyzer display:

ADDRESS	VARIABLE	LENGTH	FORMAT	R/W	NOTE
0	Modbus address	1	Signed	R/W	1-247
1	Speed	1	Signed	R/W	1: 1200 bps 2: 2400 bps 3: 4800 bps 4: 9600 bps
2	Spreading factor	1	Signed	R/W	6-12
3	Operating frequency	1	Signed	R/W	0-45
4	Parity	1	Signed	R/W	High byte: 0: none 1: even 2: odd Low byte: 0: 1 stop bit 1: 2 stop bit
7	Display sleep mode	1	Signed	R/W	Time in seconds
8	Password	1	Signed	R/W	
15	Current probe.	1	Signed	R/W	Multiplier of 100. Example: 1 = 100A 2 = 200A ... 8 = 800A
17-19	Time	1	Signed	R/W	Date, time (second, minute, hour, day, month, year)

The following table shows the Modbus addressing of the MB30 analyzer variables for reading parameters:

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32768	Phase 1 voltage	2	Float	R	V
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32774	Phase 1-2 voltage	2	Float	R	
32776	Phase 2-3 voltage	2	Float	R	
32778	Phase 3-1 voltage	2	Float	R	A
32780	Phase 1 current	2	Float	R	
32782	Phase 2 current	2	Float	R	
32784	Phase 3 current	2	Float	R	kW
32788	Phase 1 active power	2	Float	R	
32790	Phase 2 active power	2	Float	R	
32792	Phase 3 active power	2	Float	R	
32794	Total active power	2	Float	R	kVar
32796	Phase 1 reactive power	2	Float	R	
32798	Phase 2 reactive power	2	Float	R	
32800	Phase 3 reactive power	2	Float	R	
32802	Total reactive power	2	Float	R	

32804	Phase 1 apparent power	2	Float	R	kVA
32806	Phase 2 apparent power	2	Float	R	
32808	Phase 3 apparent power	2	Float	R	
32810	Total apparent power	2	Float	R	
32812	Phase 1 power factor	2	Float	R	
32814	Phase 2 power factor	2	Float	R	
32816	Phase 3 power factor	2	Float	R	
32818	Total power factor	2	Float	R	
32820	Frequency	2	Float	R	Hz
34942	Total active energy	2	Float	R	kWh
34944	Active energy consumed	2	Float	R	
34946	Active energy generated	2	Float	R	
34948	Total reactive energy	2	Float	R	kVArh
34950	Reactive energy consumed	2	Float	R	
34952	Reactive energy generated	2	Float	R	

For more technical information, additional datasheets or integration support, visit www.satel-iberia.com